Higher-order thinking questions (Optional)

Salt Preparation & Reacting Masses

1. This question is about **Salt Preparation + Reacting Masses**. (Difficulty: ⭐⭐⭐)

<u>Forensic Analysis of a Hit-And-Run Accident</u>

A hit-and-run accident has occurred late at night. The driver fled the scene, but their vehicle left behind a significant paint scrape on the victim's car. The police have identified a suspect's car of the same model and color. To prove the suspect's car was at the scene, forensic scientists need to compare the chemical composition of the paint from the scrape with a sample taken from the suspect's car.

Modern car paints often contain complex compounds for color and durability. A key pigment identified in both paint samples is a compound with the formula $Pb_xCr_yO_z$. This pigment is analyzed and found to have 64.0% of lead and 16.2% of chromium by mass.

(a) Using the percentage composition data provided, determine the empirical formula of the pigment. Show all your working clearly.

The forensic lab takes a 2.50 g sample of the pure pigment extracted from the suspect's car. Since $PbCrO_4$ is insoluble in water, the lab first reacts the sample with excess sodium carbonate solution to convert all the chromium into soluble sodium chromate (Na_2CrO_4). The resulting solution is treated with an excess of silver nitrate solution, $AgNO_3(aq)$, to precipitate out silver chromate (Ag_2CrO_4).

(b) Explain why the lab does not react lead(II) chromate directly with silver nitrate solution.

(c) Calculate the theoretical yield of silver chromate (Ag_2CrO_4) that could be precipitated from the 2.50 g sample of the pigment. (Assume silver chromate to be completely insoluble in water)

(d) The actual mass of silver chromate obtained from the experiment was 2.82 g.

(i) Calculate the percentage yield of the precipitation reaction based on your theoretical value from part (b).

(ii) Give one possible reason why the yield is not exactly 100%, and one separate reason why it might be higher than 100% if the sample were contaminated.

2. This question is about **Salt Preparation + Reacting Masses**. (Difficulty: ⭐⭐⭐⭐)

<u>Determination of Percentage by Mass of a Substance in an Impure Mixture</u>

You are given a white solid mixture consisting of lead(II) nitrate and sodium nitrate.
You are provided with common chemicals and apparatus found in a school laboratory, describe the experimental procedures required for determining the percentage by mass of lead(II) nitrate in the sample. Show the calculation steps as well.

3. This question is about **Reacting Masses**. (Difficulty: ⭐⭐⭐⭐)

Historical Development of Atomic Mass Standards and the ^{12}C Scale

Before 1960, chemists and physicists used different reference standards for atomic masses:

Chemists used naturally occurring oxygen (a mixture of ^{16}O, ^{17}O, and ^{18}O) and defined its relative atomic mass as exactly 16.
Physicists used the pure isotope oxygen-16 (^{16}O) and defined its relative atomic mass as exactly 16.

This meant the two communities had slightly different values for the Avogadro constant and for the relative atomic masses of all other elements, causing confusion in precise scientific work.

(a) On the chemist's scale, the relative atomic mass of natural oxygen (a mixture) is defined as exactly 16. Is the relative atomic mass of a pure oxygen-16 atom on this scale greater than, less than, or equal to 16? Explain.

(b) On the physicist's scale, the relative atomic mass of pure oxygen-16 is defined as exactly 16. Is the relative atomic mass of natural oxygen on this scale greater than, less than, or equal to 16? Explain.

(c) Why was having two different scales problematic for the scientific community? Give one reason.

In 1960, a new unified definition was adopted, and have been used as a standard reference till now:
"The mole is the amount of substance that contains as many elementary entities as there are atoms in exactly 12 grams of carbon-12."

(d) Give a reason why carbon-12 was chosen as the new standard instead of oxygen-16.
(Hint: How would this choice affect existing relative atomic mass data?
What physical properties make carbon suitable so that scientists can easily measure its atomic mass?)

Under this new definition of mole and the modern reference standard (carbon-12 scale):

(e) What is the molar mass of carbon-12?
(f) Write an expression for the Avogadro constant in terms of the mass of one carbon-12 atom.
(g) Explain why the Avogadro constant has the unit mol^{-1} (per mole).

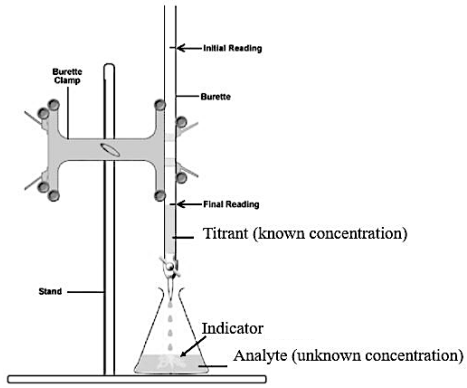
An atom of sulphur-32 is approximately 2.67 times heavier than an atom of carbon-12.

(h) Without referring to the Periodic Table, determine the mass of one mole of sulphur-32 atoms.

4. This question is about **Salt Preparation**. (Difficulty: ⭐⭐⭐⭐⭐)

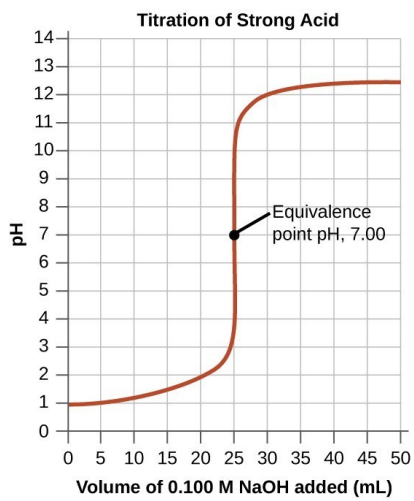
Titration Curve

A titration set-up is used to prepare $NaNO_3$ crystals. The titration set-up is shown below:



(a) The color changes from colorless to pink at the end point. What are the identities of the titrant, analyte and color indicator?

(b) The pH curve for the titration is given below:



(i) The end point is different from the equivalence point, but the end point is an indication of reaching the equivalence point.
Describe the differences between the two points. Then, based on the pH curve, explain why the end point is an indication of reaching the equivalence point.

(ii) Given that pOH = -log [OH^-(aq)] and pH + pOH = 14.

Explain the shape of the graph.
(Why is the change in pH so slow initially at high pH range? Near the equivalence point, why does the pH change rapidly? Why does the change in pH become slow again at low pH range?)